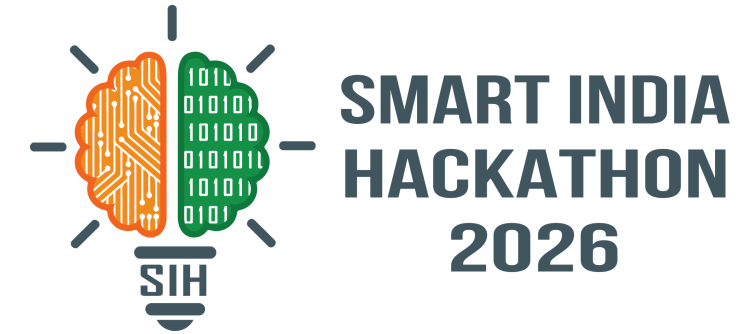




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TITLE PAGE

- **Problem Statement ID – SIH26073**
- **Problem Statement Title- AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)**
- **Theme- Disaster Management**
- **PS Category - Software**
- **Team ID - 121182**
- **Team Name- TASK TITANS**



SkyGuard AI: *Intelligent Real-Time Anomaly Detection for Automatic Weather Stations*

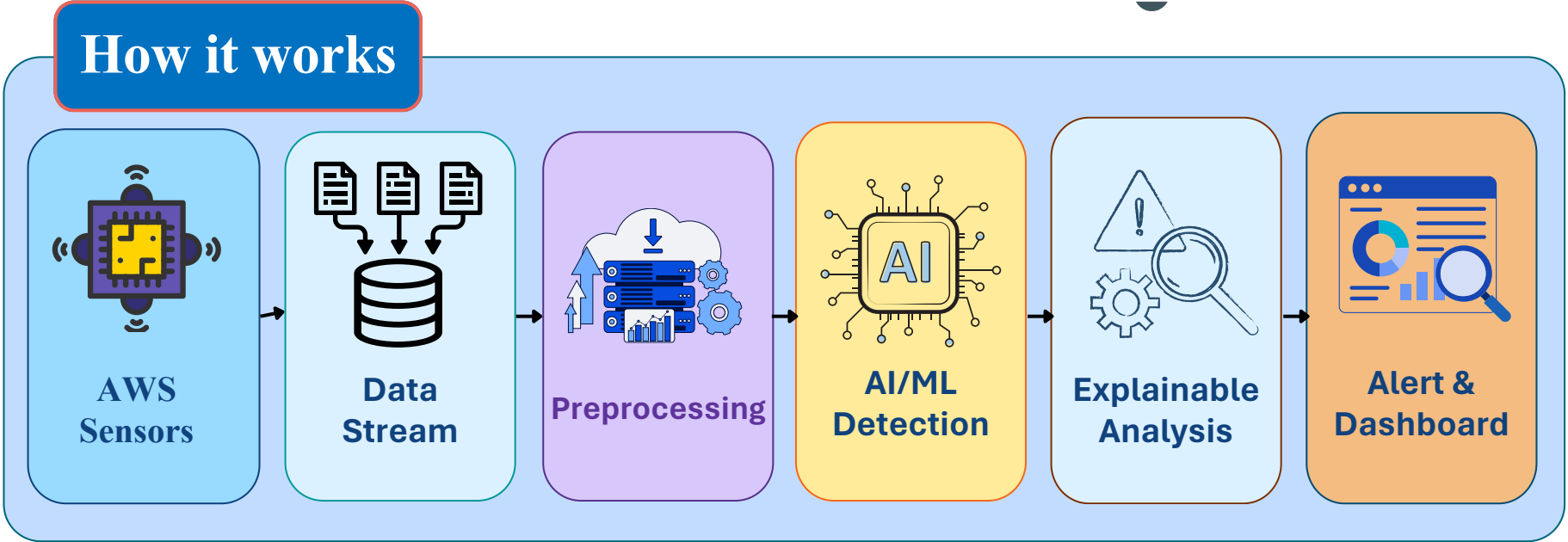


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» PROPOSED SOLUTION

SkyGuard AI is an intelligent real-time monitoring system that detects faulty, abnormal, or inconsistent observations from Automatic Weather Stations using:

Near-real-time pipelines triggered by weather alerts



Real-Time Data Ingestion

Collect Temperature, Pressure and Humidity data from live weather APIs and historical AWS datasets.

AI-Based Anomaly Detection

Use Isolation Forest and temporal/statistical models to identify spikes, frozen values, drift, missing data and inconsistent readings.

Multivariate & Spatial Analysis:

Analyze relationships between Temperature–Pressure–Humidity and compare observations across nearby stations to distinguish sensor faults from genuine weather events.

Anomaly Fusion & Classification:

Combine multiple detection scores to determine anomaly severity, confidence and probable root cause.

Explainable AI

Use SHAP to show why a reading was flagged and which parameters contributed most.

Self-Healing Data

Estimate and suggest corrected values for faulty observations while preserving the original raw data.

Sensor Health & Predictive Maintenance

Continuously calculate sensor health scores and identify degradation trends requiring maintenance.

Interactive Dashboard

Provide real-time graphs, alerts, anomaly history, station health, AI explanations and corrected-data visualization.



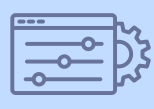


TECHNOLOGY STACK



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 **AI/ML : Python • Scikit-learn • SHAP**

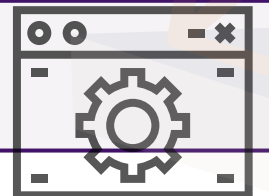
 **Data Processing: Pandas • NumPy • SciPy**

 **Backend: FastAPI • Uvicorn • Pydantic**

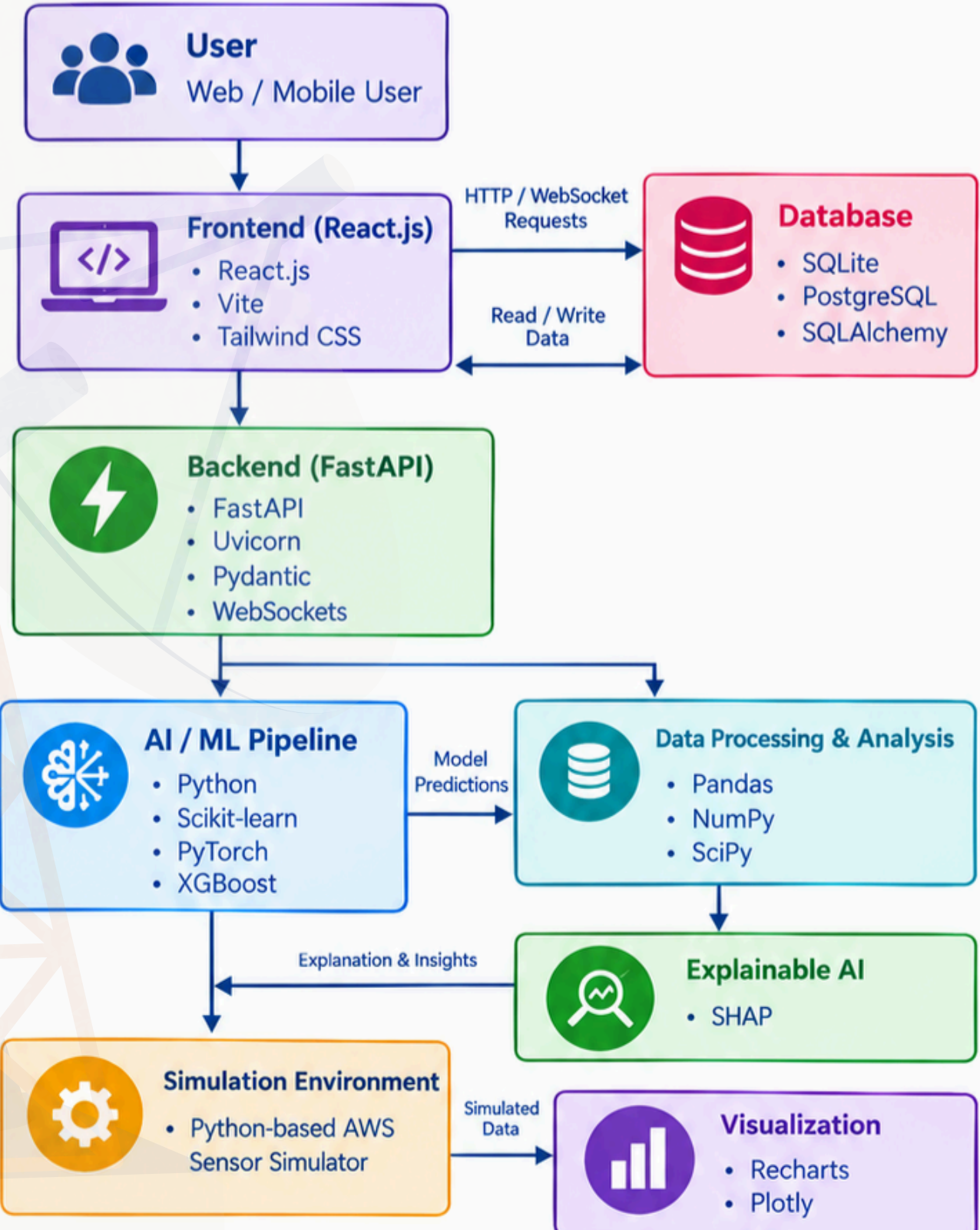
 **Real-Time: WebSockets**

 **Frontend: React.js • Vite • Tailwind CSS**

 **Database: SQLite • PostgreSQL • SQLAlchemy**

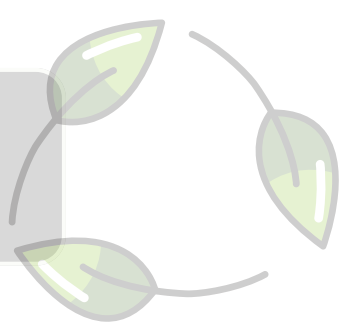
Version Control: Git • GitHub 

TECHNOLOGY FLOW





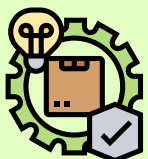
FEASIBILITY AND VIABILITY



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Feasibility

Technically Feasible



- Proven AI/ML (Python, Scikit-learn, SHAP)
- Uses only Temperature, Pressure & Humidity Data

Cost & Resource Feasible



- Open-source software
- No-extra hardware needed for prototype

Operationally Feasible



- Simple setup and easy to maintain
- Can be scaled later

Challenges & Risks



False Alarms

Hard to distinguish real weather from sensor faults



Data Quality

Missing, noisy, imbalanced or insufficient data can reduce accuracy.



Real-Time Scalability

Processing data from many AWS stations needs efficient models and reliable infrastructure.

Strategies to Overcome Challenges



Reduce False Alarms

Use temporal, multivariate and sensor-behavior analysis (not just fixed thresholds).



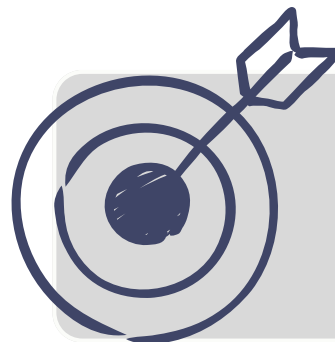
Improve Data Quality

Use preprocessing, handle missing values, simulate anomalies and retain models.



Ensure Scalability

Use lightweight ML models, FastAPI, WebSockets and modular architecture.



THE RESULT



Technically feasible



Cost effective



Privacy focused



Scalable solution

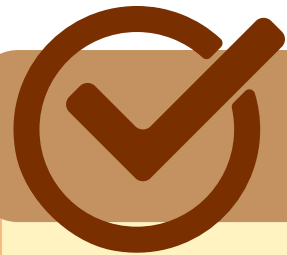


IMPACTS & BENIFITS



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Enabling AI-powered weather anomaly detection for reliable and resilient weather monitoring.



BENEFITS



Reliable Weather Data

Ensures accurate and consistent readings from AWS



Faster Fault Detection

Detects sensor issues in real time



Fewer False Alarms

Distinguishes real events from sensor faults.



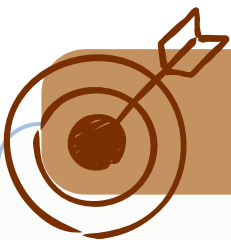
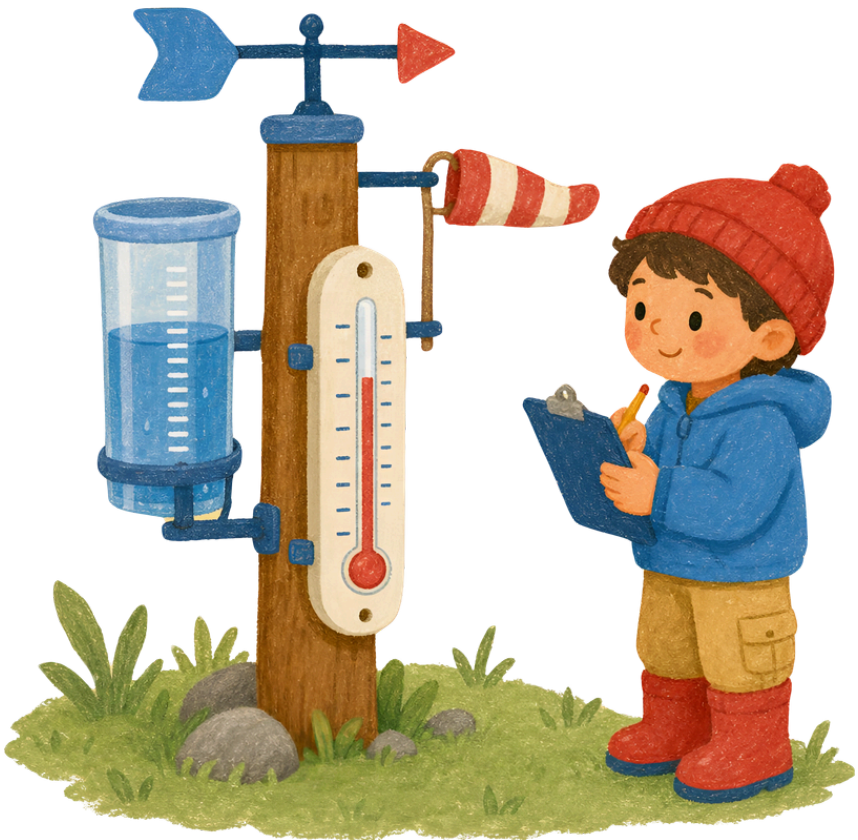
Timely Maintenance

Helps fix problems before bigger faults.



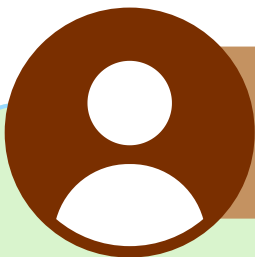
Supports Disaster Preparedness

Provides trusted data for early warnings and better decisions.



OVERALL IMPACT

- Improves sensor reliability
- Reduces maintenance downtime
- Supports accurate weather forecasting
- Strengthens disaster response



USER SCEANARIO



Faults Occurs

A sensor shows an abnormal reading (e.g., temperature spike)



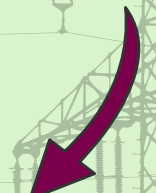
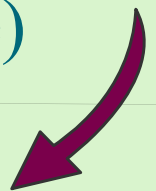
System Detects

AI/ML identifies the anomaly and checks for real weather events vs. sensor fault.



Operator Responds

Gets alert, views explanation and takes action.





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RESEARCH AND REFERENCES



1. World Meteorological Organization (WMO) — [Guide to Instruments and Methods of Observation \(WMO-No. 8\) — standards and guidance for meteorological measurements and observation quality control.](#)
2. Scikit-learn Documentation — [Novelty and Outlier Detection / Isolation Forest — basis for ML-based anomaly and outlier detection.](#)
3. Lundberg, S. M. & Lee, S.-I. (2017) — [A Unified Approach to Interpreting Model Predictions, NeurIPS — foundation for SHAP-based explainable AI.](#)
4. Liu, F. T., Ting, K. M. & Zhou, Z.-H. (2008) — [Isolation Forest, IEEE International Conference on Data Mining — foundational research for the Isolation Forest anomaly-detection method.](#)
5. WMO Instruments and Methods of Observation Programme (IMOP) — guidance on standardized observation methods, instrumentation, and quality-control procedures.

Prototype : skyguard7.netlify.app